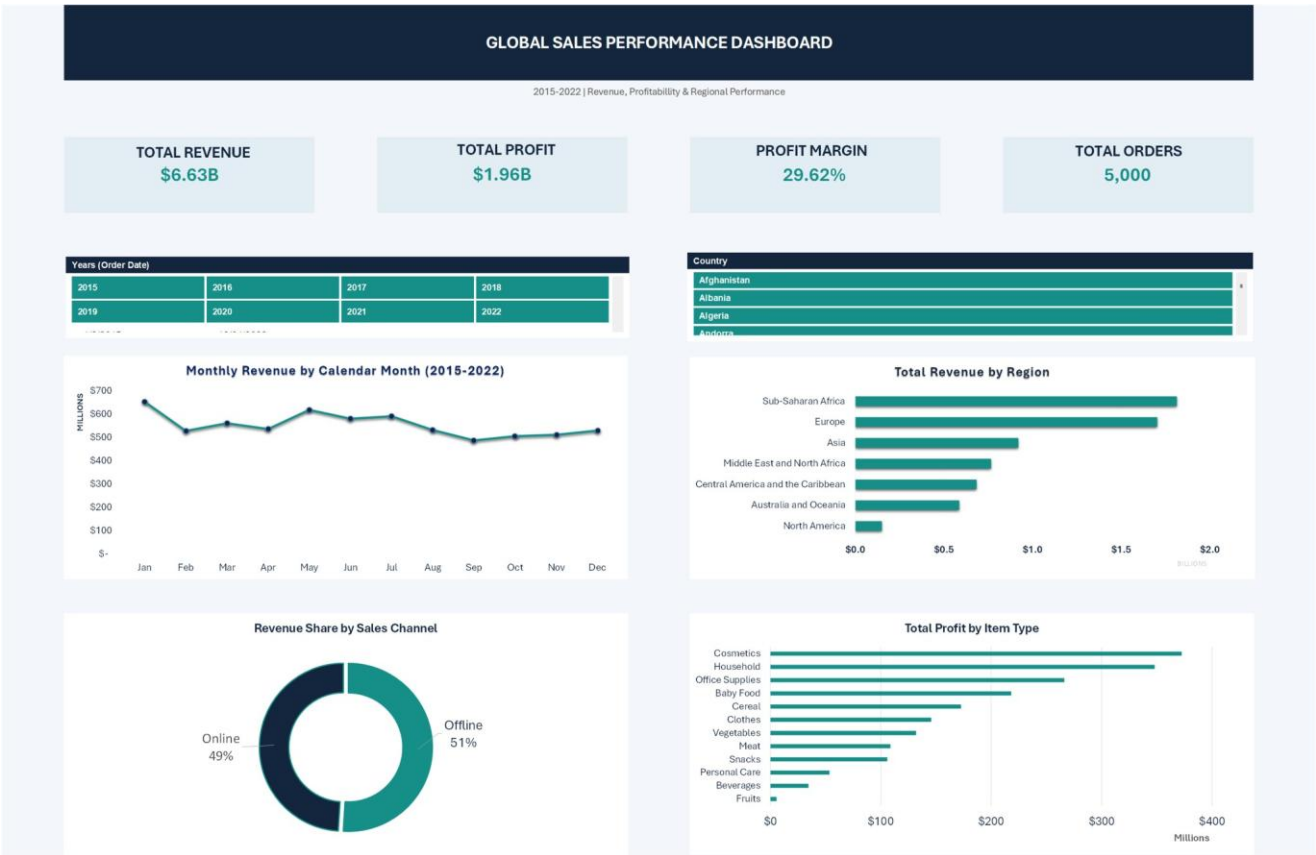
# GLOBAL SALES PERFORMANCE DASHBOARD

*Excel Analytics Portfolio Case Study | 2015–2022*

An interactive sales performance analysis using Microsoft Excel, PivotTables, PivotCharts, formulas, and slicers.




Dashboard image supplied during the project. All figures below describe the unfiltered, all-record view unless stated otherwise.

## 1. Executive Summary

This project transforms a multi-year sales dataset into an interactive Excel dashboard to examine revenue, profitability, order activity, regional performance, sales-channel contribution, and product-category profit. Users can filter the charts and KPI cards by year and country. The analysis emphasizes transparent interpretation and documents unresolved data-quality questions rather than silently excluding records.

## 2. Business Questions

- How much revenue and profit did the recorded sales generate?
- How does revenue vary by calendar month and region?
- What share of revenue comes from online versus offline channels?
- Which item types contribute the most total profit?
- How can stakeholders explore the same measures by year and country?

## 3. Dataset and Scope

| Attribute | Description |
|---|---|
| **Time period** | 2015–2022 |
| **Records** | 5,000 order records |
| **Source** | Excel sales dataset; structured table named Sales_Data |
| **Relevant fields** | Order ID, Order Date, Ship Date, Country, Region, Sales Channel, Item Type, Order Priority, Total Revenue, Total Profit |
| **Scope** | All records retained; no verified transaction-status exclusion applied |

The source workbook was analyzed in Excel. This report reflects the completed dashboard workflow and values established during the project; it is not an independent audit of accounting records.

## 4. Tools and Methods

- Excel Tables and structured references for source-data calculations.
- PivotTables to aggregate revenue, profit, and order counts.
- PivotCharts: monthly revenue line, regional revenue horizontal bars, sales-channel donut, and item-type profit horizontal bars.
- Year and Country slicers connected to all four chart PivotTables and a dedicated KPI PivotTable.
- Linked display cells and dashboard text boxes to make all four KPI cards responsive to slicers.
- Custom formatting and consistent navy, teal, and pale-blue visual design.

### KPI definitions

| KPI | Definition | Unfiltered result |
|---|---|---|
| **Total Revenue** | Sum of Total Revenue | $6,628,689,208.54 (~$6.63B) |
| **Total Profit** | Sum of Total Profit | $1,963,223,229.09 (~$1.96B) |
| **Profit Margin** | Total Profit / Total Revenue | 29.62% |
| **Total Orders** | Count of nonblank Order IDs | 5,000 |

Order IDs were checked as unique in the project, so Count of Order ID is equivalent to the number of orders. The margin is a ratio of aggregated profit to aggregated revenue, not an average of individual row margins.

## 5. Dashboard Structure

- Header: Global Sales Performance Dashboard, 2015–2022.
- KPI row: Revenue, Profit, Profit Margin, Total Orders.
- Interactive filters: Year and Country; both affect the KPI cards and all four charts.
- Top chart row: Revenue by Calendar Month (aggregated across all years) and Revenue by Region.
- Bottom chart row: Revenue Share by Sales Channel and Total Profit by Item Type.

The monthly chart groups January records together across all eight years, February records together, and so on. It is not a continuous 96-month time series, and it cannot by itself prove the same seasonal pattern in every year.

## 6. Findings and Business Interpretation

### 6.1 Calendar-month revenue

January shows the highest aggregated revenue and September the lowest in the dashboard view. This suggests differences across calendar months, but the pattern needs to be checked separately by year before concluding that it is recurring seasonality.

Recommendation: investigate campaigns, demand, pricing, product mix, and geographic contributions behind the January result; assess September performance and compare the same months year by year.

### 6.2 Regional revenue

Sub-Saharan Africa contributes the highest total revenue, followed by Europe; North America is lowest in the displayed regional ranking. Revenue alone does not establish which region is most profitable.

Recommendation: compare total profit, profit margin, product mix, and costs by region; investigate drivers of strong revenue before adapting strategies elsewhere.

### 6.3 Sales channels

Offline sales contribute approximately 51% of revenue and online sales approximately 49%. The roughly two-percentage-point difference does not justify shifting investment to offline sales on its own.

Recommendation: compare channel-specific profitability, acquisition and operating costs, retention, repeat purchasing, and customer lifetime value. CRM information would be useful, but is not established by this dataset.

### 6.4 Profit by item type

Cosmetics has the highest total profit, approximately $372.41 million, followed by Household and Office Supplies in the dashboard ranking. The top profit contributor is not necessarily the category with the highest profit margin or strongest future growth.

Recommendation: compare unit economics, profit margins, sales volumes, trends, and inventory requirements before reallocating investment. Evaluate whether successful tactics can transfer to other item types.

## 7. Data Quality, Assumptions, and Limitations

| Issue | Observed result | Decision / implication |
|---|---|---|
| **"Cancelled" in Order Priority** | 1,174 records | Retained: field is not verified as transaction status; avoid asserting these were refunds or invalid sales. |
| **Ship Date before Order Date** | 203 records | Flagged, not deleted; requires source-system investigation. |
| **Order IDs** | 5,000 unique IDs | Count used for total orders. |
| **Profit definition** | Source includes Total Profit | Profit margin reflects supplied profit; completeness of overhead and operating costs not verified. |
| **Time aggregation** | Jan–Dec across 2015–2022 | Do not describe the monthly chart as a continuous multi-year trend. |

No confirmed refunds, actual operating-cost allocations, customer-level CRM metrics, or causal explanations were established from the dashboard. These are follow-up analysis opportunities, not findings.

## 8. Recommended Next Analyses

- Validate the meaning of Order Priority = Cancelled with the data owner and, if warranted, publish a documented sensitivity comparison.
- Investigate the 203 negative shipping intervals and identify whether source dates need correction.
- Create year-by-year monthly trend and year-over-year growth analyses.
- Compare regional and channel profit margins, not revenue alone.
- Evaluate item-type profit margins, units sold, and growth over time.
- Integrate CRM and operating-cost data where available to support investment decisions.

## 9. Skills Demonstrated

Excel data validation; structured references; PivotTables and PivotCharts; KPI logic; slicer connections and dynamic dashboard measures; number formatting; dashboard design; business interpretation; communication of uncertainty and limitations.

## 10. Portfolio Deliverables

Suggested repository assets: GSPP.xlsx (working interactive dashboard), dashboard image (Excel Dashboard.jpg or exported PNG), README.md (project overview), and this case study. Remove personal or confidential data before publication and ensure the workbook is authorized for sharing.

Prepared as a portfolio case study from the dashboard development and analytical review conducted in this project.